

Course Code: SE303
Course Name: Software Engineering
Semester: Fall 2026

Course Objectives:

Understand software development processes, design principles, and project management.

UNIT 1: Software Process Models

- Waterfall
- Agile
- Spiral Model
- Scrum

UNIT 2: Requirements Engineering

- Requirement Analysis
- SRS
- Functional Requirements
- Non-functional Requirements

UNIT 3: Software Design

- UML Diagrams
- Design Patterns
- Architecture

UNIT 4: Testing

- Unit Testing
- Integration Testing
- System Testing

UNIT 5: Project Management

- Risk Management
- Estimation
- Version Control
- DevOps

Suggested Books:

- Ian Sommerville – Software Engineering